

ThaiEditBench: A Mechanism-Typed, Length-Stratified Benchmark for Thai Orthographic Error Correction

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Abstract

Thai is written without word boundaries and layers tone marks, silent consonants, and heavy loanword transliteration onto a Brahmic script. This produces orthographic error mechanisms unlike those targeted by English grammatical error correction (GEC) or Chinese spelling check (CSC). We present **ThaiEditBench**, a character-level Thai orthographic error-correction benchmark with three components: (i) a **six-class mechanism taxonomy** plus an etymology tag; (ii) **three length tiers** (sentence, paragraph, and page) that isolate robustness to long, mostly-clean context; and (iii) **contamination-robust construction**, in which naturally attested misspellings are *injected* into novel clean text screened by a memorization probe. Gold corrections are exact by construction, and a blind two-annotator study shows that the taxonomy can be applied reliably (Krippendorff’s $\alpha = 0.75$). Grading is at the character-cluster level with span-based matching adapted from ERRANT. Evaluating **16 models**, we find that (1) under the benchmark’s conditions, models that locate an error almost always fix it correctly, so what separates them is *detection*; (2) the evaluated Thai-specialized models do not lead the strongest general models; and (3) long, mostly-correct context exposes **over-edit accumulation** and clean-text fidelity failures that sentence-level evaluation hides. On real errors mined from Wikipedia edit history, scores are somewhat lower but the model ranking transfers (Spearman $\rho = 0.92$).

1 Introduction

Thai orthography is error-prone in script-specific ways: tone-mark selection and placement, silent-consonant marks, homophonous consonants, vowel-form confusions, consonant-cluster metathesis, and inconsistent loanword transliteration. These are **character-level** errors that turn on **specific mechanisms**, and Thai’s lack of whitespace word boundaries makes them hard to localize and score with

token-based tools. Yet the resources to study them are missing. Grammatical-error-correction benchmarks are English-centric (Bryant et al., 2017, 2019; Napoles et al., 2017), and the closest character-level analogue is Chinese spelling check (Tseng et al., 2015; Liu et al., 2011). For Thai, existing error-annotated resources are social-media, informal, or learner-domain (Lertpiya et al., 2020; Nakwijit and Purver, 2022), and there is no formal-register, mechanism-typed, contamination-controlled orthographic benchmark. Document-level and context-aware GEC study correction over long context (Yuan and Bryant, 2021), but no benchmark isolates length and error-sparsity robustness for Thai orthographic correction.

Contributions. (1) **ThaiEditBench**, a character-level Thai orthographic correction benchmark with a six-class mechanism taxonomy plus an etymology tag and **three length tiers** (sentence, paragraph, page). It is **contamination-robust by construction**: naturally attested misspellings are injected into novel clean text screened by a memorization probe, so gold corrections are exact. A blind annotation study measures how reliably the taxonomy can be applied (Krippendorff’s $\alpha = 0.75$). (2) An **ERRANT-style orthographic edit-typer for Thai** that aligns edits at TCC level and labels each with one of the six mechanism classes, enabling **per-mechanism span-F0.5** (cf. ChERRANT’s dual-granularity scoring (Zhang et al., 2022)). We use it as a scoring and diagnosis instrument, not as a general Thai grammatical error classifier. (3) A **16-model evaluation**: models that find an error almost always fix it, so detection separates them; the evaluated Thai-specialized systems do not lead; weaker models accumulate over-edits under long context; and the ranking transfers to naturally occurring errors (Spearman $\rho = 0.92$). We release code, data, grader, edit-typer, and the variant map at <https://github.com/yongsa-nut/ThaiEditBench>.

2 Related Work

GEC and CSC. English grammatical error correction is evaluated with span-based metrics. The M^2 scorer matches a system’s edits against gold edit spans (Dahlmeier and Ng, 2012), ERRANT (the ERRor ANnotation Toolkit) additionally infers an error type for each edit automatically and reports an F0.5 score (Bryant et al., 2017), and GLEU measures fluency (Napoles et al., 2015). ERRANT-style span matching aligns the wrong and corrected text into a minimal set of edit spans, then credits a system for each gold span it recovers with the correct replacement; we adapt this scheme to Thai TCC units (Section 4.2). Character-level Chinese spelling check is the closest task, scoring position-based detection and correction F1 without word boundaries (Tseng et al., 2015; Wu et al., 2013; Yu et al., 2014), with confusion-set construction (Liu et al., 2011) and the MuCGEC/ChERRANT dual-granularity scorer (Zhang et al., 2022). ERRANT has been ported to Arabic (Belkebir and Habash, 2021) and Korean (Yoon et al., 2023), but not Thai. Wikipedia-revision mining of typo pairs is established for English (Grundkiewicz and Junczys-Dowmunt, 2014) and Japanese (Tanaka et al., 2020).

Thai NLP. We build on PyThaiNLP for tokenization and dictionary tooling (Phatthiyaphaibun et al., 2023) and on the Thai Character Cluster as the minimal orthographic unit (Theeramunkong et al., 2000). Prior Thai spelling correction targets social text (Lertpiya et al., 2020, 2023), and related work studies the semantics of Thai misspellings (Nakwijit and Purver, 2022). The VISTEC-TP-TH-2021 Twitter corpus includes misspelling detection and correction as well as word-segmentation annotation, but it is informal Twitter text rather than a formal-register, mechanism-typed orthographic benchmark (Limkonchotiwat et al., 2021).

3 The ThaiEditBench Benchmark

3.1 Task

Given a Thai passage containing character-level orthographic errors, return the passage with those errors corrected and **nothing else changed**: no rewriting, re-spacing, numeral, or punctuation edits. We score under a strict **minimal-edit contract**, where any change to a non-erroneous span is an *over-edit* and is penalized.

3.2 Mechanism taxonomy

We define six orthographic mechanism classes, each carrying an **etymology tag**: (1) tone mark, (2) consonant-pair, (3) silent-consonant, (4) vowel, (5) cluster or metathesis, and (6) loanword transliteration. Worked examples per class are in Appendix A. Canonical forms are verified against the Royal Institute Dictionary (ORST). Class 6 is governed by a **loanword-absorption rule**: only *contemporary* foreign loans are class 6, while long-integrated borrowings (Tamil, Khmer, Persian, and Sanskrit–Pali doublets) are treated as native.

3.3 Construction

- **(a) Confusion-set injection.** The confusion set holds 1,044 wrong→correct pairs across the six classes. Every wrong form is a **naturally attested misspelling**, harvested from the Thai Wiktionary appendix of commonly misspelled words and from fixes made by a Thai Wikipedia spelling-cleanup bot; a rule-based classifier assigns each pair a mechanism class (96.5% agreement with 90 hand-labeled pairs; Appendix C). Only the placement is synthetic: pairs are injected into *novel clean text* by **whole-token-run matching** (not substring), which protects short valid forms, and each injected error records its Thai Character Cluster (TCC) span, class, and etymology. Applying the gold edits restores the clean original for 100% of items, so the gold is exact.
- **(b) Cleanup-bot mining.** Real errors in their original sentences are mined from the same bot’s edit history, keeping only fixes whose corrected word is listed in the ORST dictionary and whose wrong and correct forms align cleanly, following the revision-mining tradition (Tanaka et al., 2020; Grundkiewicz and Junczys-Dowmunt, 2014), with the bot’s fix as gold. These items form a separate natural-error evaluation set (Section 5.5).

Registers. The clean text spans news (articles from the ThaiSum corpus), encyclopedic (Thai Wikipedia), and government and legal (the Royal Gazette and Thai government press releases), all permissively redistributable.

3.4 Contamination control

The benchmark reduces contamination risk by evaluating systems on **corrupted passages that do not**

appear verbatim in the source corpora. A residual risk remains: a model that memorized a clean source passage could reconstruct it directly instead of detecting and correcting the injected errors. We therefore run a **verbatim-continuation memorization probe** on the source passages and place the lowest-memorization material in a separate held-out test split.

3.5 Length tiers

The same pipeline produces three tiers that vary passage length and error sparsity. **T1** (sentence) has 795 items of ~ 100 characters each; **T2** (paragraph) has 223 items of ~ 424 characters; and **T3** (page) has 124 items of $\sim 1,242$ characters, with 974, 276, and 193 gold edits respectively. Injection is stratified by class quota rather than uniform: the class mix follows the skew of the attested pairs (the per-class share of T1 gold edits correlates with that of the attested pairs at Pearson $r = 0.80$) and is partially flattened so that every class has at least 115 gold edits at T1 (Appendix C). The error budget is **dense at the sentence level**, with roughly one error per 100 characters, and deliberately **sparse at paragraph and page length**, where most sentences are clean distractors. T2 and T3 therefore measure **precision under long correct context** (over-editing), not just recall, an axis the CSC and GEC literature does not test.

3.6 Quality control

Because errors are introduced by construction, the gold corrections are exact. We add two automatic guards.

- **Cross-model consensus audit.** When at least $k=3$ of the 16 systems produce the *same* normalized output that differs from gold, we flag the item. We drop it only when that output fixes a residual typo in the *source* text itself, and keep *shared misses* (the consensus equals the corrupted input) and *systematic over-edits* on punctuation or register. This deterministic audit removed 24 of 819 (T1), 27 of 250 (T2), and 26 of 150 (T3) items (Appendix B).
- **Variant reconciliation.** Spellings with two accepted forms are reconciled to the gold form before scoring, token-aware and compound-safe. The map has four equivalence groups (eight word forms): an orthographic doublet, two loanword transliterations, and an informal-to-formal pronoun pair. It is gold-directional,

so it can accept an attested variant but cannot turn a wrong answer into a right one, and it changes 2.35% of model outputs.

Orthographic correction is essentially **single-reference**, since there is one correct spelling, so we do not adopt multi-reference GEC machinery.

3.7 Human validation of the labels

The mechanism classes are assigned automatically. To measure how reliably the taxonomy can be applied, two native Thai speakers independently labeled 60 items, blind to any machine label: the 40 items of a curated high-confidence subset of the mined items, plus synthetic items that bring each class to ten. For each marked span they judged whether it is a genuine error and assigned a class. Both raters judged all 60 spans to be genuine errors, and their class agreement reaches Krippendorff’s $\alpha = 0.752$ (Krippendorff, 2011). Where the raters agree (48 items), their class matches the automatic label on 35. Of the 25 items on which any two of the three labels differ, 21 fall at two documented conventions: the loanword-absorption rule (17) and the restriction of class 2 to homophone groups containing an Indic-origin letter (4). Appendix E gives the per-class results.

4 Evaluation Protocol

4.1 Metrics

We grade at the TCC level (Theeramunkong et al., 2000) with ERRANT-style span matching (Bryant et al., 2017). The **primary metric is correction F1**: minimal TCC edits are derived by aligning the wrong text against *both* the gold and the system output under the **same** Levenshtein alignment, so an output identical to the reference scores exactly 1.0 (SIGHAN-CSC-comparable (Tseng et al., 2015)), and we report it with a 95% **item-bootstrap confidence interval** (Appendix D). **Detection F1** credits the correct error site and ignores the replacement. **Clean-sentence fidelity** is the fraction of error-free sentences left untouched, the exact and construction-grounded precision signal for T2 and T3. The edit-typer (Section 4.2) also gives an ERRANT-style **span-F0.5**, overall and per mechanism. Differences between systems are tested with a **paired item-level bootstrap** (Koehn, 2004) over 10,000 resamples. Word-level GLEU is released with the data as a supplementary metric only, because Thai word segmentation is unreliable for this task.

4.2 ERRANT-style edit-typer

We adapt the ERRANT/ChERRANT design (Bryant et al., 2017; Zhang et al., 2022) to Thai orthography. For a (wrong, correct) pair: extract edit spans at **TCC level** (the same alignment the grader uses), expand each to its enclosing word (PyThaiNLP `newmm` (Phatthiyaphaibun et al., 2023)), and label it with one of the six mechanism classes (plus an edit-operation tag); class 6 is etymology-gated by a loan lexicon. Typing both the gold and the system edits enables **per-mechanism span-F0.5** on any (wrong, correct) pair, without construction labels. On the constructed items, the typer recovers 98.9% of the gold edit spans and agrees with the construction class on 98.1% of them. Its inventory is orthographic-mechanism-specific and reuses the construction-time classifier, so we treat it as a **scoring instrument**, not a general Thai grammatical error classifier.

4.3 Models and inference

We evaluate 16 models, each in a single fixed configuration, at temperature 0 where supported, with one fixed Thai instruction prompt. The **frontier proprietary** models are GPT-5.5 (medium and high reasoning), Gemini 3.5 Flash, Claude Opus 4.7 (non-thinking), Claude Sonnet 4.6 (non-thinking), and GPT-5.4-mini. The **open-weight** models are Gemma-4-31B, DeepSeek V4 Pro and Flash, GLM-5.1, MiniMax-M2.7, and Qwen3.6-35B-A3B. The **Thai-native** models are Typhoon 2.5 (30B) and three ThaiLLM 8B-instruct models served through the national ThaiLLM gateway, namely OpenThaiGPT (v7.2), Typhoon-S, and THaLLE (Pipatanakul et al., 2024; Pipatanakul and Taveekitworachai, 2026). A prompt-sensitivity ablation is in Appendix G.

5 Results and Analysis

5.1 Leaderboard

Table 1 reports correction F1 on the three tiers and on the natural-error set (post-audit $n = 795/223/124/79$); the full table is in Appendix D. The frontier clusters at 0.95 to 0.97 on T1, and the confidence intervals separate three strata: the frontier and the strongest open-weight models, then GLM-5.1, then the Thai-native models, MiniMax, and Qwen. Within the frontier, T1 differences are mostly not significant: GPT-5.5 (high) is tied with GPT-5.5 (med) and Claude Opus 4.7. The length-

model	T1	T2	T3	T3 fid	Nat.
GPT-5.5 (high)	0.973	0.969	0.855	0.974	0.943
GPT-5.5 (med)	0.969	0.955	0.856	0.971	0.931
Claude Opus 4.7	0.968	0.948	0.848	0.974	0.892
Gemini 3.5 Flash	0.965	0.960	0.819	0.960	0.865
Gemma-4-31B	0.950	0.954	0.889	0.981	0.942
GPT-5.4-mini	0.949	0.917	0.853	0.988	0.917
Claude Sonnet 4.6	0.947	0.906	0.824	0.972	0.882
DeepSeek V4 Flash	0.947	0.945	0.828	0.966	0.841
DeepSeek V4 Pro	0.926	0.926	0.746	0.954	0.818
GLM-5.1	0.846	0.812	0.665	0.954	0.759
Typhoon-S 8B [†]	0.782	0.702	0.619	0.965	0.687
THaLLE 8B [†]	0.749	0.595	0.517	0.829	0.727
Typhoon 2.5 30B [†]	0.742	0.631	0.467	0.896	0.740
OpenThaiGPT v7.2 [†]	0.741	0.477	0.454	0.896	0.717
MiniMax-M2.7	0.740	0.720	0.470	0.968	0.702
Qwen3.6-35B-A3B	0.705	0.383	0.308	0.836	0.567

Table 1: Correction F1 by tier and on the natural-error set (Nat.), with T3 clean-sentence fidelity. **Bold** = column best; underlined = not separable from the best by a paired bootstrap test ($p \geq 0.05$). Paired tests account for item-level correlation, so a lower score can be tied with the best while a higher one is not. [†] = Thai-native.

degradation slope (T1 to T3) is the sharpest discriminator (Section 5.3).

General frontier models dominate, while Thai-specialized models trail. Typhoon 2.5 scores 0.742 on T1 and 0.467 on T3, and the 8B national-gateway models score 0.74 to 0.78 on T1 but fall to 0.45 to 0.62 on T3. The strongest **open-weight** model is a general 31B model, Gemma-4. The frontier leads it on T1 ($\Delta = 0.022$ against GPT-5.5 high, $p < 0.001$), and on T3 it has the highest point estimate (0.889) but is statistically tied with the frontier (against GPT-5.5 med, $\Delta = +0.034$, 95% CI $[-0.015, 0.081]$). Under this setup the evaluated Thai-specialized systems do not lead, and the gap widens with length (Gemma-4 against Typhoon 2.5 on T3: $\Delta = 0.423$, $p < 0.001$). This comparison is confounded by differences in model scale, serving stack, and inference configuration, so it is not evidence about Thai-specialized pretraining in general.

5.2 Where models fail: finding versus fixing

We decompose every gold edit into **fixed**, **mis-corrected** (the system edits exactly the gold span but with a different replacement), and **missed** (no system edit matches the gold span) (Table 2). Across all 16 models, the miscorrected share never exceeds 0.036 on T1, while missed edits account for nearly all failures. The pattern holds within every mechanism class, including the hardest (loanword: mis-

model	fixed	miscorrected	missed
GPT-5.5 (high)	0.993	0.000	0.007
Gemma-4-31B	0.956	0.004	0.040
Qwen3.6-35B-A3B	0.817	0.018	0.165
Typhoon 2.5 30B	0.764	0.036	0.201
All 16, T1	0.866	0.008	0.126
All 16, T3	0.828	0.005	0.167
All 16, natural	0.804	0.007	0.189

Table 2: Gold-edit outcomes (T1 unless stated). The miscorrected share stays near zero throughout.

corrected 0.003 against missed 0.189), at both the dense T1 and the sparse T2 and T3, and on natural errors (Section 5.5). Because a miscorrected edit requires an exact span match, a model that rewrites a wider region around an error is scored as missed; such touched-but-unmatched edits are only 1.3 to 1.8% of gold edits per tier, so even counting them as wrong fixes leaves untouched errors five times more frequent. Under this benchmark’s conditions (formal register, single-reference gold, and one or two errors per passage), models rarely fail by choosing the wrong spelling once they locate an error, so what the leaderboard separates is **detection**. Spelling knowledge still matters, because recognizing a misspelling presupposes knowing the correct form; the evaluated models largely have that knowledge and differ in applying it to find errors. Whether this holds for dense, interacting errors is open (Limitations).

5.3 Length robustness

Sentence-tier rank does **not** predict paragraph or page behaviour, and the **degradation slope from T1 to T3 is the sharpest discriminator** (Figure 1a). The frontier loses only 0.06 to 0.15 correction F1 (GPT-5.5-med 0.969→0.856; Gemma-4 0.950→0.889), while weak models fall off a cliff (Typhoon 2.5 0.742→0.467; Qwen 0.705→**0.308**, a drop of 0.40). We measure over-editing per 10,000 clean TCCs, which normalizes for exposure. This rate is roughly flat or falls across tiers (Qwen 105→107→65; Typhoon 2.5 57→33→27; Figure 1b), so over-editing accumulates with length through exposure rather than through a rising false-positive rate. The discriminating signal is the *level* of that rate, not its slope: weak models over-edit at 5 to 20 times the frontier’s per-unit rate, which stays below ~15 per 10,000 clean TCCs at every tier. Clean-sentence **fidelity** agrees: the frontier holds at or above 0.95, while the heaviest over-editors sit

at 0.78 to 0.90. **Over-editing and missing are two faces of the same weak discrimination**: a model that cannot tell correct from incorrect both misses real errors and rewrites correct text. The exception is **Gemini 3.5 Flash**, which fixes 0.985 of gold edits at T3 yet drops to 0.819 correction F1 on precision, the one frontier model limited by over-editing rather than by detection.

5.4 Per-mechanism difficulty

Table 7 (Appendix D) breaks the fixed rate down by class and tier. At T1, class 6 (loanword) is the hardest mechanism, with 0.81 of edits fixed against 0.86 to 0.90 for the other classes, and the failure is again **missed**, not miscorrected. The class spread narrows with length, from 0.09 at T1 to 0.05 at T3, while the overall fixed rate falls from 0.87 to 0.83, so the length effect is largely mechanism-independent. Within class 6 the ranking does **not** favour Thai-native models, suggesting that the difficulty lies in the mechanism, not in cross-lingual knowledge. Register agrees: loanword-dense encyclopedic text is hardest (recall 0.71 to 0.82, against 0.76 to 0.90 for news and government).

5.5 Natural errors

We evaluate all 16 systems on the mined cleanup-bot items, each a real error in its original encyclopedic sentence. The consensus audit of Section 3.6 removes 24 of the 103 items: in 20, most systems also fix a second, unmarked error in the sentence, and in 4 the recorded bot fix is itself incomplete (Appendix F). On the remaining 79 items, mean correction F1 is 0.81 against 0.87 on T1, the system ranking transfers (Spearman $\rho = 0.92$ with T1; Table 1), and the decomposition reproduces (Table 2). Without the audit, mean F1 is 0.72 and the miscorrected share is 0.04 to 0.07 for every system, almost entirely on those gold defects, while the ranking still transfers ($\rho = 0.88$; Appendix F).

6 Conclusion

ThaiEditBench is a mechanism-typed, length-stratified, contamination-robust benchmark for Thai orthographic correction, with an ERRANT-style edit-typer for per-mechanism scoring. Across 16 models, found errors are almost always fixed, so the leaderboard separates models by **detection**; the evaluated **Thai-specialized systems do not lead**; weaker models show **over-edit accumulation** under long context; and the ranking transfers to natural errors.

Limitations

Sparse error density (primary). The long tiers (T2 and T3) are deliberately sparse, with 1 to 3 errors scattered across otherwise-clean passages, to isolate precision under long correct context. We therefore do **not** test **high error-density** regimes such as heavy-typo input, degraded OCR, or L2 writing, where errors co-occur, interact, mask one another, and correction order matters. Extending ThaiEditBench to dense multi-error passages, and testing whether the detection finding survives when errors crowd each other, is the most important next step.

A fixed inventory of attested errors. Every injected error comes from 1,044 attested pairs, each mapping one wrong form to one correct form. The benchmark therefore does not cover misspellings outside the inventory, and a system that consulted the released inventory would be answering a different, tool-augmented question; all systems are evaluated zero-shot without it. The class mix follows the distribution of attested misspelling types, not the frequency of errors in running text.

Scope of the natural-error check. The audited natural set has 79 items from one cleanup bot in one register, with no item typed as class 6 and only five as class 5, so it supports the ranking and the decomposition but not per-class conclusions; loan-word behaviour is measured on injected errors only. The audit relies on cross-model consensus, and the bot’s fix is taken as gold, so the natural gold is less certain than the construction gold of the tiers. A larger, naturally distributed test set, which the released mining pipeline supports, is the main way to strengthen this check.

Label reliability. The annotation study covers 60 items and two raters. Per-class agreement on ten items per class is noisy (α from 0.49 to 1.00) and lowest for class 6, whose boundary depends on the absorption convention, and the automatic class matches the rater consensus on 35 of 48 items. Per-mechanism results near the class 6 and class 2 boundaries should be read with this uncertainty.

Configured-system comparison. Each system is evaluated under one fixed configuration and one fixed Thai prompt. Reasoning modes, token budgets, provider defaults, serving stacks, and model scale are not controlled, so the leaderboard reflects configured-system behaviour rather than an intrinsic ranking of base-model capability. A prompt ablation on two systems shifts correction F1 by at

most 0.07 without reordering them (Appendix G), but we do not sweep prompts across all systems.

Minimal-edit vs fluency boundary. Occasionally a frontier “over-edit” is actually catching a residual error that the injected gold did not mark, which slightly understates the best models.

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A Mechanism taxonomy in detail

Each class is defined by the orthographic *mechanism* of the fault, not its surface character. Multiple worked examples per class:

- **1 tone mark:** คี๋→คะ, นี๋→นะ, จี๋จั่น→จิกจั่น, กระทั่ง→กระทั่ง
- **2 consonant-pair:** กฎหมาย→กฎ หมาย, กบฏ→กบฏ, อาภาส→อาภาศ
- **3 silent-consonant:** กลยุทธ์→กล ยุทธ์, ขาดดุลย์→ขาดดุล, เทห์→เท
- **4 vowel:** สังเกต→สัง เกต, มาตรฐาน→มาตรฐาน, ฉะเพาะ→เฉพา
- **5 cluster/metathesis:** กระทั่งหัน→กะทั่งหัน, กระบังลม→กะบังลม
- **6 loanword:** กราฟฟิค→กราฟิก, กงสุล→กงสุล, ลิ้งค์→ลิงก์

Class 2 scope. Class 2 covers swaps within a homophone group that contains an Indic-origin letter, for example ศ/ษ/ส, ท/ธ, and น/ณ. Swaps between letters outside these groups, such as ฟ/พ, fall in class 5 together with other structural edits.

Loanword-absorption rule. Only *contemporary* foreign loans, chiefly modern English technical and commercial terms (กราฟิก, ดิจิทัล, ลิงก์), are class 6. Long-integrated borrowings are treated as **native** even though etymologically foreign: Portuguese (กะละแม), Burmese (กะปี้), Mon (กะพง), Khmer (กบฏ), and Tamil, Persian, and Sanskrit–Pali doublets. Borderline cases (e.g. กงสุล, *consul*) are resolved by ORST listing + Wiktionary borrowed-term categories, and the tag is recorded per item so the rule is auditable.

B Gold quality control and the consensus audit

Why construction-gold still needs a cleanup pass. Injection guarantees that applying the gold edits recovers the clean original (verified 100%), so the *injected* error’s correction is exact. It does **not** guarantee that the **source passage was typo-free**: the permissive corpora contain a small rate of residual orthographic errors. When an item’s source carries such a residual typo, the gold inherits it, so a model that fixes *that* genuinely wrong span is unfairly scored as a miss or an over-edit. We remove these automatically.

Cross-model consensus audit. For every item we pool all 16 outputs (variant-reconciled) and look for a *consensus* output, the same normalized string from at least $k=3$ systems, that differs from gold. The discrimination is three-way. (i) If the consensus equals the corrupted input, this is a *shared miss*, the gold is correct, and we **keep** the item. (ii) If the consensus is a punctuation, whitespace, or register variant, this is a systematic *over-edit*, the precision signal we want, and we **keep** it. (iii) If the consensus is a *different* spelling fix at an unmarked span, this is a residual *source* typo, and we **drop** it. Only case (iii) is a gold defect, and the explicit test of whether the consensus equals the input prevents conflating it with shared misses and over-pruning. The audit removed 24 (T1), 27 (T2), and 26 (T3) items, for example ทั้งสิ้น→ทั้งสิ้น, ปฏิเศษ→ปฏิเสธ, and โดบ→โดย, all residual *source* typos. **Post-audit invariant:** no item in any tier is missed by all 16 models. This is the signature that residual gold defects are gone while genuinely hard items remain. Every drop follows deterministically from the three-way rule above, and we release a self-contained HTML view of the audit (gold-versus-consensus diffs labeled drop, keep, or variant) for inspection.

Variant map. The four equivalence groups are สิทธิ/สิทธิ์, ดิจิตอล/ดิจิทัล, ออร์แกนิก/ออร์แกนิก, and the informal pronoun คำ/เขา. Words in which the final mark is mandatory, such as สิทธิสิทธิ์, are protected from reconciliation.

C Construction details

Confusion set. Wrong→correct pairs were harvested from the Thai Wiktionary appendix of commonly misspelled words and from the fix history of a Thai Wikipedia spelling-cleanup bot, then deduplicated and cleaned (fragments, multi-token and spacing-only pairs, and context-dependent real-word pairs removed), giving 1,044 canonical pairs. A rule-based classifier assigns the mechanism class from the character difference; it agrees with 90 hand-labeled bot-fix pairs on 96.5%. Etymology is assigned from the Wiktionary borrowed-term categories under the absorption rule, which moves contemporary loans into class 6. Canonical forms are verified against the offline ORST word list distributed with PyThaiNLP (87.5%; the misses are modern loanwords absent from that list), and each pair is also tagged with its edit operation. Table 4 gives the per-class sizes.

Injection and mining. Injection uses **whole-**

tier	unit	items	chars	err/it	news/gov/enc
T1 sent.	1 sent.	795	100	1.23	275/368/152
T2 para.	≈5 sent.	223	424	1.24	80/79/64
T3 page	≈15 sent.	124	1,242	1.56	42/43/39

Table 3: Per-tier statistics (post-audit; Section 3.6). The tiers are 100% confusion-set injection; the cleanup-bot items form the separate natural-error set.

class	pairs	T1	T2	T3
1 tone mark	43	163	42	34
2 consonant-pair	168	137	36	24
3 silent-consonant	177	166	54	26
4 vowel	189	182	60	39
5 cluster/metathesis	338	211	55	53
6 loanword	129	115	29	17
total	1,044	974	276	193

Table 4: Confusion-set pairs and gold edits per class and tier. The class shares of T1 gold edits correlate with those of the attested pairs at Pearson $r = 0.80$ (Spearman 0.82); cluster/metathesis is the largest class in both.

token-run matching so the 51 valid short forms (≤ 3 characters) that coincide with a confusion target are never corrupted. Cleanup-bot mining reads the bot’s contribution history on Thai Wikipedia and keeps a fix only when the corrected word is listed in ORST and the wrong and correct forms align without shifting neighbouring characters; the mined items are kept out of the three tiers. The curated subset used in the annotation study keeps at most two contexts per distinct error and drops fragments of two characters or fewer. A verbatim-continuation **memorization probe** (Typhoon 2.5 30B) finds little memorization for every source: the mean longest common prefix between the model’s continuation and the true text is 0.003 to 0.017 of its length. The lowest-memorization text forms the held-out test split.

Release and license. The injected items are released as train (249), dev (606), and public-test (819 before the audit) splits, plus a held-out test split (868) that this paper does not evaluate; T1 is the audited public-test split. We release the benchmark for research use under CC BY-SA (data) and the MIT license (code), consistent with the permissive redistribution terms of the source corpora.

D Full results

Pooled per-class outcome split (T1, all 16 models).

class	gold edits	fixed	miscorrected	missed
1 tone	2,608	0.896	0.011	0.092
2 consonant	2,192	0.876	0.006	0.117
3 silent-consonant	2,640	0.887	0.010	0.103
4 vowel	2,896	0.857	0.002	0.140
5 cluster	3,392	0.863	0.012	0.126
6 loanword	1,968	0.808	0.003	0.189

Over-edits per 100 items across tiers (absolute accumulation).

model	T1	T2	T3
GPT-5.5 (med)	7	8	41
Gemma-4-31B	6	8	32
GLM-5.1	10	23	106
Typhoon 2.5 30B	32	82	201
OpenThaiGPT v7.2	20	110	206
Qwen3.6-35B-A3B	59	268	484

Over-edit rate per 10,000 clean TCCs (length-normalized; the Figure 1b data). The per-unit rate is roughly flat or falls across tiers, so the per-item growth above reflects text exposure, not a rising false-positive rate.

model	T1	T2	T3
GPT-5.5 (med)	12.3	3.0	5.5
Gemma-4-31B	11.4	3.0	4.3
GLM-5.1	17.8	9.3	14.3
Typhoon 2.5 30B	56.8	32.6	27.0
OpenThaiGPT v7.2	35.0	44.0	27.8
Qwen3.6-35B-A3B	105.4	106.8	65.1

Register effect (pooled correction recall, T1/T2/T3). encyclopedic 0.82/0.79/0.71; news 0.87/0.85/0.76; government 0.89/0.90/0.83. Per-model $\times$ class correction recall and span-F0.5 per type are released with the data.

E Annotation study

Design. The 60 items comprise the 40 curated mined items (Appendix C) and 20 synthetic items chosen so that each automatic class has ten items. Each item shows one marked span with its proposed correction. Raters answer two questions per item: whether the marked span is a genuine orthographic error, and which of the six classes it belongs to, following a written guideline with the class definitions and the two conventions of Appendix A. No automatic label is shown. The two raters, one of them an author, worked independently. Etymology and edit-operation tags are not annotated, since they are a dictionary lookup and a deterministic function of the edit.

Results. Error validity: both raters accepted all 60 spans. Class agreement: Krippendorff’s $\alpha = 0.752$ (nominal, $n = 60$). Table 8 gives one-vs-rest α and raw agreement per automatic class; with ten items per class these values are noisy.

model	T1 det	T1 corr	T1 span-F0.5	T2 corr	T2 fid	T3 corr	T3 fid
GPT-5.5 (high)	0.973	0.973	0.961	0.969	0.990	0.855	0.974
GPT-5.5 (med)	0.969	0.969	0.956	0.955	0.983	0.856	0.971
Claude Opus 4.7	0.971	0.968	0.956	0.948	0.986	0.848	0.974
Gemini 3.5 Flash	0.969	0.965	0.951	0.960	0.983	0.819	0.960
Gemma-4-31B	0.955	0.950	0.947	0.954	0.988	0.889	0.981
GPT-5.4-mini	0.959	0.949	0.951	0.917	0.986	0.853	0.988
Claude Sonnet 4.6	0.952	0.947	0.938	0.906	0.977	0.824	0.972
DeepSeek V4 Flash	0.952	0.947	0.939	0.945	0.986	0.828	0.966
DeepSeek V4 Pro	0.928	0.926	0.910	0.926	0.970	0.746	0.954
GLM-5.1	0.851	0.846	0.879	0.812	0.961	0.665	0.954
Typhoon-S 8B [†]	0.788	0.782	0.841	0.702	0.961	0.619	0.965
THaLLE 8B [†]	0.758	0.749	0.787	0.595	0.775	0.517	0.829
Typhoon 2.5 30B [†]	0.776	0.742	0.728	0.631	0.881	0.467	0.896
OpenThaiGPT v7.2 [†]	0.754	0.741	0.776	0.477	0.887	0.454	0.896
MiniMax-M2.7	0.747	0.740	0.801	0.720	0.974	0.470	0.968
Qwen3.6-35B-A3B	0.721	0.705	0.652	0.383	0.776	0.308	0.836

Table 5: Full leaderboard: detection and correction F1, span-F0.5 from the edit-typer, and clean-sentence fidelity across the three length tiers. **Bold** = column best; [†] = Thai-native. Table 1 marks which systems are statistically tied with each column’s best. Table 6 gives the confidence intervals. Per-class recall and per-type span-F0.5 are released with the data.

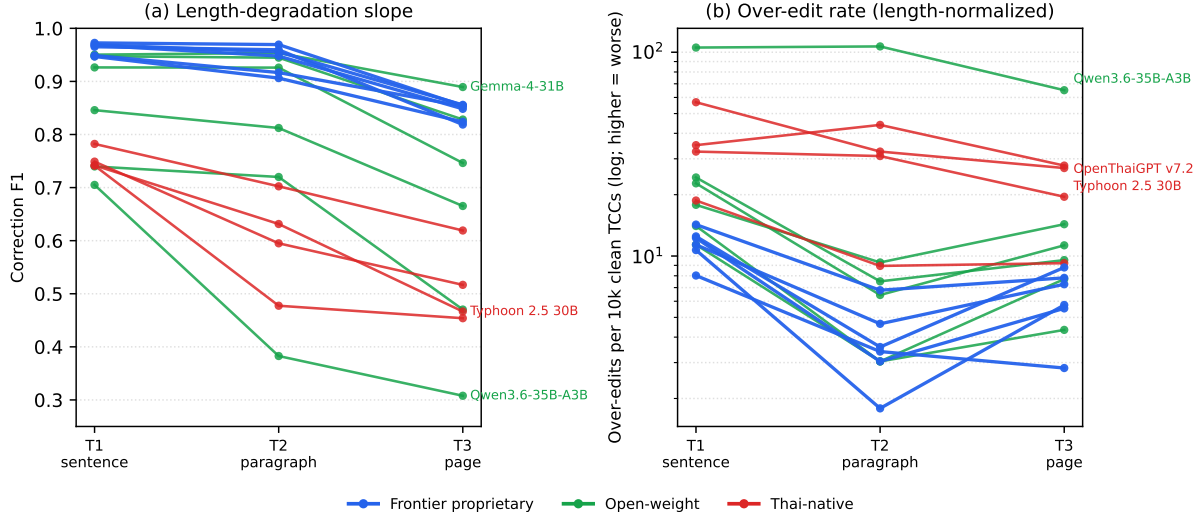


Figure 1: Length robustness. (a) Correction-F1 slope from T1 to T3. The frontier (blue) and the strongest open-weight model (Gemma-4, green) stay high while weak models slope off. (b) Over-edit rate per 10,000 clean TCCs on a log scale, where higher is worse. The rate is roughly flat across length, so the per-passage accumulation (the absolute counts in this appendix) reflects text exposure rather than a rising false-positive rate. Weak models over-edit at many times the frontier’s per-unit rate at every tier.

Where labels differ. The raters split on 12 items and agree with each other against the automatic label on 13. Of these 25, 17 involve the loanword-absorption rule in one of two directions: raters labeled the surface mechanism of a contemporary loan (for example, a consonant swap in เวียดนาม→เวียดนาม rated class 2 rather than 6), or recognized a loan that the loan lexicon missed (เยอรมันนี่→เยอรมนี, typed class 5 but rated 6); old borrowings such as เวทมนตร์ split the raters between class 3 and 6. Four more are homophone

swaps outside the Indic-origin groups (คน→คน three times, and กบฏ→กบฏ), which the taxonomy places in class 5 and both raters placed in class 2. The remaining four concern whether a stray or misplaced diacritic counts as a tone-mark, vowel, or silent-consonant error. The disagreements therefore sit at the class boundaries fixed by convention rather than at error perception.

model	T1	T2	T3
GPT-5.5 (high)	[0.964, 0.980]	[0.952, 0.984]	[0.803, 0.902]
GPT-5.5 (med)	[0.960, 0.977]	[0.934, 0.974]	[0.806, 0.902]
Claude Opus 4.7	[0.958, 0.978]	[0.924, 0.970]	[0.778, 0.911]
Gemini 3.5 Flash	[0.955, 0.975]	[0.942, 0.976]	[0.759, 0.874]
Gemma-4-31B	[0.938, 0.962]	[0.931, 0.974]	[0.846, 0.928]
GPT-5.4-mini	[0.936, 0.960]	[0.888, 0.944]	[0.810, 0.893]
Claude Sonnet 4.6	[0.933, 0.959]	[0.875, 0.936]	[0.753, 0.890]
DeepSeek V4 Flash	[0.935, 0.959]	[0.918, 0.968]	[0.775, 0.880]
DeepSeek V4 Pro	[0.912, 0.939]	[0.902, 0.949]	[0.684, 0.806]
GLM-5.1	[0.823, 0.867]	[0.767, 0.855]	[0.545, 0.781]
Typhoon-S 8B [†]	[0.759, 0.805]	[0.652, 0.751]	[0.552, 0.683]
THaLLE 8B [†]	[0.715, 0.780]	[0.540, 0.649]	[0.436, 0.598]
Typhoon 2.5 30B [†]	[0.716, 0.767]	[0.580, 0.684]	[0.399, 0.538]
OpenThaiGPT v7.2 [†]	[0.701, 0.777]	[0.386, 0.571]	[0.380, 0.534]
MiniMax-M2.7	[0.714, 0.765]	[0.671, 0.767]	[0.392, 0.552]
Qwen3.6-35B-A3B	[0.673, 0.736]	[0.308, 0.470]	[0.227, 0.419]

Table 6: 95% item-bootstrap confidence intervals on correction F1 (10,000 resamples). [†] = Thai-native.

class	T1	T2	T3
1 tone mark	0.896	0.868	0.807
2 consonant-pair	0.876	0.885	0.831
3 silent-consonant	0.887	0.865	0.800
4 vowel	0.857	0.870	0.854
5 cluster/metathesis	0.863	0.807	0.838
6 loanword	0.808	0.802	0.819
all classes	0.866	0.851	0.828
spread (max-min)	0.088	0.083	0.054

Table 7: Fraction of gold edits fixed, by mechanism class and tier, pooled over all 16 models. Gold-edit counts per cell are in Table 4.

automatic class	α (one-vs-rest)	agreement
1 tone mark	0.793	1.00
2 consonant-pair	0.921	0.90
3 silent-consonant	0.611	0.60
4 vowel	0.773	0.70
5 cluster/metathesis	1.000	1.00
6 loanword	0.485	0.60
all (multiclass)	0.752	0.80

Table 8: Inter-rater agreement per automatic class (ten items each).

F Natural-error evaluation

The natural set comes from the edit history of a Thai Wikipedia spelling-cleanup bot: 103 items, each a real error in its original encyclopedic sentence, with the bot’s fix as gold. We apply the consensus audit of Section 3.6 with the same threshold ($k = 3$). It drops an item when the consensus output makes a spelling change the gold does not, and keeps it when the consensus equals the input, differs only in whitespace or punctuation, or is a partial fix on the way to the gold. It removes 24 items: in 20

the consensus also fixes a second, unmarked error elsewhere in the sentence (for example เพี้ย→เพีย), and in 4 the recorded fix is itself incomplete (ตำ recorded as ตำ, น้ำ as น้ำ, and สร้าง as สร้าง, where the intended words are ตำ, น้ำ, and สร้าง). The audited set has 79 items and 80 gold edits (automatic classes 1: 24, 2: 19, 3: 13, 4: 18, 5: 5). Table 9 gives per-model results on the audited set, with correction F1 and the miscorrected share on all 103 items for comparison. On the audited set the ranking correlates with the T1 ranking at Spearman $\rho = 0.92$ (Pearson $r = 0.93$); on all 103 items, at $\rho = 0.88$ ($r = 0.93$).

G Prompt and inference

A single fixed Thai instruction prompt is used for all models. It asks the model to return the passage with orthographic errors corrected and **nothing else changed**, explicitly forbidding spacing, punctuation, and numeral edits. Temperature is 0 where supported. Claude models use the Anthropic Message Batches API in non-thinking mode. GPT-5.x models use the Responses API at the stated reasoning effort with a 3k-token budget. Open reasoning models run in native thinking mode with large token budgets, and token-runout counts as a failure. ThaiLLM-gateway models are reached with a browser User-Agent header, because the gateway’s web application firewall otherwise blocks the default SDK user agent.

Prompt sensitivity. We rerun the full T1 set for one strong general model (Gemma-4-31B) and one Thai 8B model (Typhoon-S) under two alternative prompts: the same instruction in English,

model	audited set ($n = 79$)						all items ($n = 103$)	
	corr F1	95% CI	det F1	fixed	miscorrected	missed	corr F1	miscorrected
GPT-5.5 (high)	0.943	[0.907, 0.975]	0.943	0.938	0.000	0.062	0.800	0.048
Gemma-4-31B	<u>0.942</u>	[0.893, 0.981]	0.955	0.912	0.013	0.075	0.819	0.058
GPT-5.5 (med)	<u>0.931</u>	[0.886, 0.969]	0.931	0.925	0.000	0.075	0.791	0.048
GPT-5.4-mini	<u>0.917</u>	[0.859, 0.969]	0.930	0.900	0.013	0.087	0.811	0.058
Claude Opus 4.7	<u>0.892</u>	[0.785, 0.968]	0.892	0.925	0.000	0.075	0.791	0.038
Claude Sonnet 4.6	0.882	[0.810, 0.942]	0.895	0.838	0.013	0.150	0.771	0.058
Gemini 3.5 Flash	<u>0.865</u>	[0.750, 0.969]	0.865	0.963	0.000	0.037	0.757	0.038
DeepSeek V4 Flash	0.841	[0.750, 0.921]	0.841	0.825	0.000	0.175	0.757	0.048
DeepSeek V4 Pro	0.818	[0.727, 0.909]	0.818	0.900	0.000	0.100	0.729	0.038
GLM-5.1	0.759	[0.663, 0.853]	0.759	0.750	0.000	0.250	0.689	0.048
Typhoon 2.5 30B [†]	0.740	[0.649, 0.831]	0.751	0.800	0.013	0.188	0.655	0.048
THaLLE 8B [†]	0.727	[0.638, 0.811]	0.727	0.650	0.000	0.350	0.670	0.048
OpenThaiGPT v7.2 [†]	0.717	[0.623, 0.806]	0.717	0.650	0.000	0.350	0.630	0.048
MiniMax-M2.7	0.702	[0.607, 0.785]	0.702	0.588	0.000	0.412	0.663	0.038
Typhoon-S 8B [†]	0.687	[0.587, 0.774]	0.702	0.562	0.013	0.425	0.626	0.038
Qwen3.6-35B-A3B	0.567	[0.468, 0.670]	0.606	0.738	0.050	0.212	0.544	0.067

Table 9: Natural-error evaluation. **Bold** = best on the audited set; underlined = not separable from the best by a paired bootstrap test ($p \geq 0.05$); [†] = Thai-native.

model	Thai (default)	English	Thai + 2 examples
Gemma-4-31B	0.950	0.924	0.885
Typhoon-S 8B	0.782	0.799	0.834

Table 10: T1 correction F1 under three prompts ($n = 795$).

and the Thai instruction with two worked examples from the train split, which is disjoint from T1 (Table 10). Prompt choice shifts correction F1 by at most 0.07, and Gemma-4 stays ahead of Typhoon-S under every prompt. The two models move in opposite directions: the worked examples help the Thai 8B model and hurt the general model.